
PHYSICS-INFORMED NEURAL NETWORKS FOR MARTIAN THERMAL DIFFUSIVITY RECOVERY AND THE ROLE OF THE MEASUREMENT MODEL

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ABSTRACT

We apply physics-informed neural networks (PINNs) to recover the thermal diffusivity of Martian subsurface soil from NASA InSight HP³ mission data, the first in-situ thermal measurements on Mars. A progression of models is developed, from a 1D point-sensor model to a 2D axisymmetric model, each addressing a limitation identified in the previous. The point-sensor model produces a 440% inconsistency between amplitude-derived and phase-derived κ values, rendering any recovered parameter physically meaningless regardless of convergence. The spatial-average model recovers $\kappa = 6.28 \times 10^{-8} \text{ m}^2/\text{s}$ with 0.49% convergence spread across three initialisations, confirmed within 1% by an independent analytical solution. The 60% discrepancy with Spohn et al.'s value is consistent with the mole's thermal fin effect, which the uniform spatial average does not capture. Inverse PINNs can converge reliably to a physically consistent but incorrect parameter value when the forward model does not match the true measurement process. These results demonstrate that for real-instrument inverse problems, the fidelity of the observation operator matters more than network architecture, loss weighting, or training strategy.

Keywords Physics-informed neural networks · Inverse heat conduction · Mars · InSight HP³ · Thermal diffusivity · Ill-posed problems

1 Introduction

The thermal diffusivity of Martian soil governs how heat propagates through the subsurface. It determines the depth and persistence of diurnal and seasonal temperature variations, which in turn control the thermal environment for potential brine stability, ice persistence, and subsurface habitability (Spohn et al., 2024). Knowledge of thermal diffusivity also constrains soil composition and density, which are relevant to geotechnical considerations for future surface operations. Despite its importance, direct measurement of this property from orbital data is not possible, as it must be inferred from in-situ temperature measurements at depth.

NASA's InSight mission provided the first such measurements on Mars, recording both the surface temperature via the RAD radiometer and the subsurface temperature via the HP³ mole, a 40 cm aluminium thermal probe spanning depths from 1.7 to 36.3 cm. Spohn et al. (2024) derived a thermal diffusivity of $\kappa = (3.93 \pm 0.39) \times 10^{-8} \text{ m}^2/\text{s}$ from these data using Fourier analysis of the diurnal temperature wave. Their analysis relies on calibration equations derived from a detailed numerical model of the mole that accounts for its spatial averaging and thermal fin effects.

Physics-informed neural networks (PINNs) offer an alternative approach to parameter inference from PDE-governed systems. By embedding the governing equation directly into the training loss, PINNs can recover unknown parameters from sparse observations without requiring a mesh or explicit numerical discretisation. Inverse PINNs have been

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demonstrated for a range of PDE systems including Navier-Stokes, Burgers, and diffusion equations (Raissi et al., 2019), with recent applications to inverse heat conduction and thermal diffusivity recovery (Cai et al., 2021; Billah et al., 2023). However, most applications use synthetic data with idealised observation operators in which the sensor is assumed to record the physical field exactly at a point. Applications to real instrument data, where the measurement process introduces its own physics, remain rare.

Recovering κ from boundary or near-surface temperature measurements is a classic example of an ill-posed inverse problem (Woodbury et al., 2023). As a thermal signal propagates through a medium, sharp features attenuate and smooth out, so that two distinct surface signals can produce nearly indistinguishable interior measurements. This violates the uniqueness and stability conditions required of a well-posed problem (Hadamard, 1923). Classical IHCP methods handle this through explicit regularisation, such as the function specification method and Tikhonov regularisation, which constrain the space of admissible solutions by penalising abrupt or unphysical variation. PINNs offer an alternative where the physics residual acts as an implicit regulariser. This approach is arguably stronger than a generic smoothness penalty, since it enforces consistency with the actual governing heat equation rather than smoothness alone. Whether this implicit, physics-based regularisation is sufficient on its own is the central question this paper investigates.

To our knowledge, this is among the first applications of PINNs to extraterrestrial thermal data, using InSight HP³ measurements to recover the thermal diffusivity of Martian regolith. A progression of models is developed: a point-sensor inverse formulation, a spatial-average formulation that models the mole as an extended sensor, and 2D axisymmetric models that attempt to capture radial heat exchange. The spatial average model recovers $\kappa = 6.28 \times 10^{-8} \text{ m}^2/\text{s}$ with 0.49% convergence spread across three initialisations, confirmed within 1% by an independent analytical solution. The 60% discrepancy with Spohn et al.’s value is consistent with the mole’s thermal fin effect, which the uniform spatial average does not capture. The central finding is that the fidelity of the observation operator (the mathematical model of how the sensor interacts with the physical field) determines whether an inverse PINN recovers a meaningful parameter, and if so, which one.

2 Physical Setup and Data

2.1 Governing equation

The temperature field in Martian regolith is governed by the one-dimensional heat diffusion equation:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2} \quad (1)$$

where $T(z, t)$ is the soil temperature at depth z and time t , and κ is the thermal diffusivity (m^2/s). Thermal diffusivity controls how rapidly temperature disturbances propagate through a medium: high κ means heat diffuses quickly and temperature signals penetrate deeper; low κ means heat diffuses slowly and signals are confined near the surface. On Mars, the diurnal surface temperature swing of $\sim 60 \text{ K}$ penetrates only a few centimetres into the soil. This is because Martian regolith has a very low thermal diffusivity.

2.2 The InSight HP³ Instrument

NASA’s InSight lander has operated on the surface of Mars since November 2018 (Banerdt et al., 2020). Its Heat Flow and Physical Properties Package (HP³; Spohn et al., 2018) includes two instruments relevant to this work: the RAD radiometer, which measures surface temperature, and the TEM-A sensor, which measures the temperature of a subsurface thermal probe known as the mole.

The mole is a 40 cm aluminium cylinder with a thermal conductivity of approximately $200 \text{ W/m}\cdot\text{K}$, buried at 30° inclination and spanning depths from $z_0 = 1.7 \text{ cm}$ to $z_1 = 36.3 \text{ cm}$. Because aluminium conducts heat roughly $5000\times$ faster than the surrounding Martian soil ($k_{\text{soil}} \approx 0.039 \text{ W/m}\cdot\text{K}$; Grott et al., 2021), the mole is nearly isothermal along its entire length. The TEM-A sensor is a 31.5 cm copper-on-Kapton foil wrapped along the hull. It records the equilibrium temperature of the aluminium body, not the soil temperature at any single depth. This distinction between what the sensor physically measures and what it is often assumed to measure becomes central to our results.

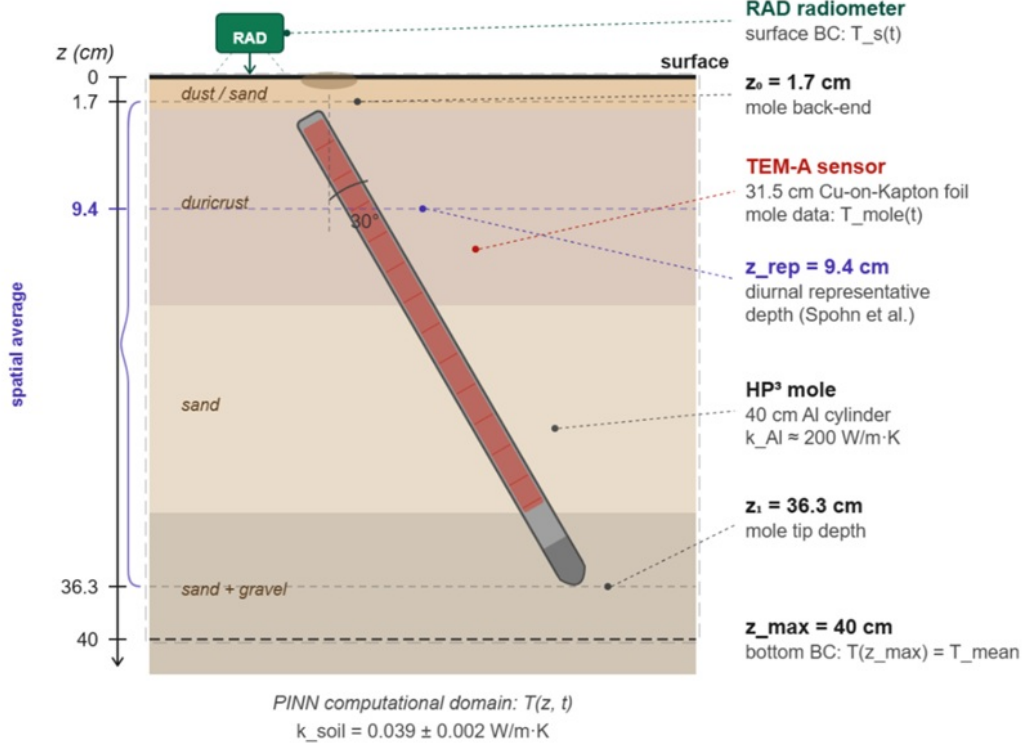


Figure 1: Cross-section of the HP³ mole in the Martian regolith, showing the PINN computational domain, sensor positions, and the spatial averaging range from z_0 to z_1 . Soil stratigraphy follows Spohn et al. (2024).

2.3 Data

We use the RAD surface temperature record from Sol 511 (InSight RAD Science Team, 2019) and the TEM-A mole temperature record from Sol 1202 (InSight HP³ Science Team, 2021). These sols were selected because Sol 511 ($L_s = 195^\circ$, MY35) and Sol 1202 ($L_s = 208.8^\circ$, MY36) fall in the same Martian month (Month 7, early northern autumn), ensuring comparable seasonal insolation despite being recorded one Mars year apart. This pairing was also adopted by Spohn et al. (2024).

The PINN is trained on a single-sol domain: $z \in [0, 0.4]$ m and $t \in [0, 88775]$ s (one Martian sol). Two boundary conditions define the problem:

- **Surface** ($z = 0$ m): the RAD diurnal temperature record, which provides the periodic thermal forcing.
- **Depth** ($z = 0.4$ m): fixed at $T = 225.6$ K, the diurnally averaged TEM-A mole temperature from Sol 1202. At this depth the diurnal signal has fully decayed and the temperature is approximately constant over a single sol.

The mole temperature data serves a different role depending on the model variant: in the forward problem it provides validation data; in the inverse problem it provides the measurement constraint from which κ is inferred.

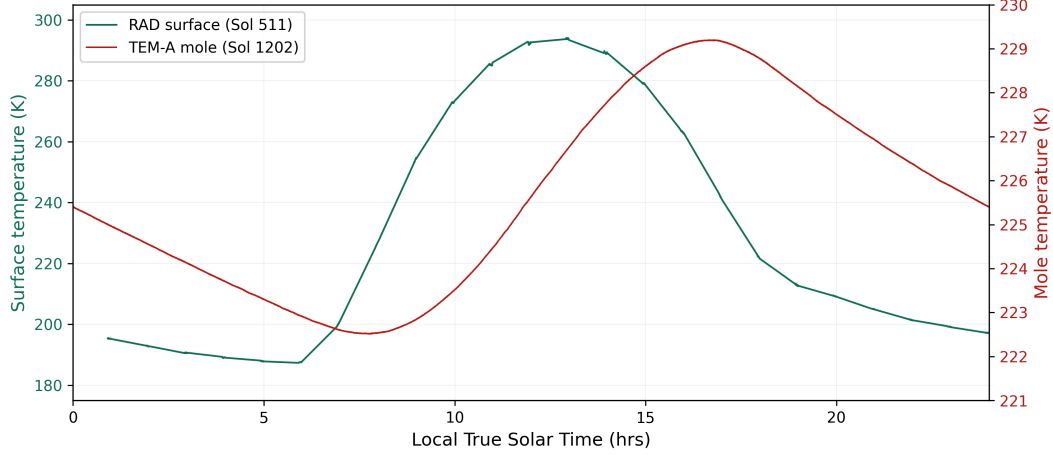


Figure 2: Diurnal temperature records used as PINN inputs: RAD surface temperature (Sol 511, left axis) and TEM-A mole temperature (Sol 1202, right axis). The amplitude attenuation from ~ 107 K at the surface to ~ 6.7 K at mole depth, and the phase lag between the two signals, are the observational signatures from which κ is recovered.

Table 1: Physical and geometric parameters used in this work.

Parameter	Symbol	Value	Source
Soil thermal conductivity	k_{soil}	0.039 ± 0.002 W/m·K	Grott et al. (2021)
Soil thermal diffusivity (reference)	κ	$(3.93 \pm 0.39) \times 10^{-8}$ m ² /s	Spohn et al. (2024)
Aluminium thermal conductivity	k_{al}	≈ 200 W/m·K	—
Mole length	L	40 cm	Spohn et al. (2024)
Mole radius	R_{mole}	1.35 cm	Spohn et al. (2024)
Mole inclination	—	30° from vertical	Spohn et al. (2024)
Mole back-end depth	z_0	1.7 cm	Spohn et al. (2024)
Mole tip depth	z_1	36.3 cm	Spohn et al. (2024)
Domain depth	z_{max}	40 cm	This work
Far-field radius (2D)	r_{max}	25 cm	This work
Sol duration	P	88,775 s	NASA/JPL
Angular frequency	ω	7.08×10^{-5} rad/s	$2\pi/P$
Diurnal skin depth	δ	3.33 cm	$\sqrt{\kappa P/\pi}$
Mean mole temperature	T_{mean}	225.6 K	TEM-A Sol 1202
Diurnal representative depth	z_{rep}	9.4 cm	Spohn et al. (2024)

2.4 Reference Value

Spohn et al. (2024) derived $\kappa = (3.93 \pm 0.39) \times 10^{-8}$ m²/s from the diurnal temperature variation by measuring the amplitude ratio and phase shift between the surface and mole signals. Their method uses calibration equations derived from a detailed thermal mathematical model of the mole that accounts for both the spatial averaging over the mole’s depth extent and its thermal fin effect. This value serves as our primary reference throughout the paper.

3 PINN Methodology

3.1 Physics-Informed Neural Networks

A standard neural network learns to map inputs to outputs by minimising the error between its predictions and training data. While effective for interpolation, such models can produce physically implausible predictions away from data points as they do not know the underlying physics.

Physics-informed neural networks (PINNs), introduced by Raissi et al. (2019), address this by incorporating the governing PDE directly into the training objective. The network must simultaneously fit the available data and satisfy

the physical equations at a set of collocation points sampled throughout the domain. This dual constraint enables PINNs to generalise physically even in data-sparse regions.

3.2 Network Architecture

The PINN takes two inputs (depth z and time t) and outputs the predicted temperature $T(z, t)$. Unless otherwise noted, all experiments use a fully connected network with 4 hidden layers of 64 neurons each and $\tanh$ activation functions, giving approximately 13,000 trainable parameters. Both inputs are normalised to $[0, 1]$ before entering the network.

Table 2: Network architecture common to all experiments.

Property	Value
Inputs	z, t (1D) or r, z, t (2D), normalised to $[0, 1]$
Output	T
Hidden layers	4
Neurons per layer	64
Activation	$\tanh$
Parameters	$\approx 13,000$
Optimiser	Adam (Optax/JAX)
κ parameterisation	$\exp(\log \kappa)$
Framework	JAX

3.3 Loss Function

The total loss is a weighted sum of three components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}} \quad (2)$$

The boundary condition loss enforces that the network matches prescribed values at the domain boundaries:

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_{\text{BC}}} \sum_{i=1}^{N_{\text{BC}}} |T_{\text{net}}(z_i, t_i) - T_{\text{BC}}(z_i, t_i)|^2 \quad (3)$$

This includes the RAD surface temperature at $z = 0$ and the fixed deep temperature $T = 225.6$ K at $z = 0.4$ m (see Section 2).

The data loss enforces agreement with the TEM-A mole measurement. In the point-sensor formulation, this is a single evaluation at the representative depth; in the spatial average formulation, it is replaced by the discrete average described in Section 4.3.

The physics loss enforces the heat equation at N_{coll} collocation points sampled randomly throughout the domain:

$$\mathcal{L}_{\text{phys}} = \frac{1}{N_{\text{coll}}} \sum_{i=1}^{N_{\text{coll}}} \left| \frac{1}{\kappa} \frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial z^2} \right|^2 \quad (4)$$

Note the rescaled form $(1/\kappa) \cdot \partial T / \partial t - \partial^2 T / \partial z^2 = 0$ rather than the standard $\partial T / \partial t - \kappa \cdot \partial^2 T / \partial z^2 = 0$. As discussed in Section 4.1, this rescaling was essential for physical correctness. The partial derivatives $\partial T / \partial t$ and $\partial^2 T / \partial z^2$ are computed exactly via automatic differentiation.

3.4 Inverse Problem: Learning κ

In the inverse formulation, the thermal diffusivity κ is treated as a learnable parameter optimised alongside the network weights. To ensure positivity and improve optimiser stability across the small magnitudes involved ($\sim 10^{-8}$), κ is parameterised in log-space:

$$\kappa = \exp(\log \kappa) \quad (5)$$

where $\log \kappa$ is the learnable parameter. This transforms the multiplicative scale of κ into an additive one, allowing the optimiser to take steps of comparable size regardless of the current value of κ .

3.5 Training

All models are implemented in JAX and trained using the Adam optimiser from the Optax library. Forward problems typically use 200,000 iterations with a piecewise learning rate schedule (1×10^{-3} for the first 120,000 iterations, decaying to 1×10^{-4} for the remainder). Inverse problems require significantly more iterations. We used up to 2 million for the point-sensor model and 350,000–900,000 for the spatial average model, to assess whether κ has genuinely stabilised. Loss weights are set to $\lambda_{\text{BC}} = 100$, $\lambda_{\text{phys}} = 1$ for all inverse experiments. The data weight differs by model: $\lambda_{\text{data}} = 50$ for the point-sensor model and $\lambda_{\text{data}} = 200$ for the spatial average model, reflecting the stronger constraint required to fit the spatially averaged signal. All experiments use $N_{\text{coll}} = 10,000$ collocation points, sampled once at the start of training from a uniform distribution over the domain.

4 Progressive Experiments

4.1 Forward Problem Validation

In the forward problem, κ is fixed at Spohn et al.’s published value of $3.93 \times 10^{-8} \text{ m}^2/\text{s}$ and the PINN learns only the temperature field $T(z, t)$. This serves two purposes: validating that the PINN can solve the heat equation with real boundary data, and establishing a baseline against which the inverse problem results can be compared.

Several standard approaches were tested to improve convergence: Fourier feature embeddings (Tancik et al., 2020), learning rate schedules, increased network capacity (up to 4×128 , $\sim 50,000$ parameters), loss weight rebalancing, and increased collocation point density (up to 50,000). Of these, learning rate scheduling provided modest improvements, while Fourier features and increased collocation density had no effect. Larger networks achieved lower data-space error but produced catastrophically unphysical depth profiles. For example, the 4×128 network achieved RMSE of 0.006 K at the mole depth while generating temperature swings exceeding ± 400 K at other depths. This demonstrated that low RMSE at measurement locations does not guarantee physical correctness.

The key improvement came from diagnosing the physics residual itself. In the standard residual formulation $\partial T / \partial t - \kappa \cdot \partial^2 T / \partial z^2$, the normalised diffusivity $\kappa_{\text{eff}} (\approx 0.022$ in normalised units) renders the curvature term nearly negligible. This allows the model to develop large spatial oscillations at minimal cost to the physics loss. We reformulated the residual as $(1/\kappa) \cdot \partial T / \partial t - \partial^2 T / \partial z^2 = 0$, equalising the contribution of both terms. After rescaling, depth profiles shifted from ± 400 K swings to the physically expected 200–260 K range. This rescaling was the single most impactful change across all experiments. This finding is consistent with Wang et al. (2022), who showed using neural tangent kernel theory that imbalanced loss components in PINNs lead to pathological training dynamics, and with Krishnapriyan et al. (2021), who demonstrated that PINN failure modes arise from the loss landscape rather than insufficient network expressivity.

The best physically-valid forward solution achieved RMSE ≈ 1.44 K at the mole depth using a 4×64 network with the rescaled residual. The remaining error shows a clean sinusoidal pattern with a consistent phase lag and amplitude mismatch between the PINN prediction and the TEM-A data. This structured error signature indicates that the fixed $\kappa = 3.93 \times 10^{-8} \text{ m}^2/\text{s}$ does not exactly reproduce the mole signal, pointing directly toward the inverse problem.

4.2 Point-sensor inverse problem

In the forward problem, $\kappa = 3.93 \times 10^{-8}$ was fixed and the PINN only tried to learn $T(z, t)$. In the inverse formulation, κ is parameterised as a learnable variable initialised at a chosen value, and the PINN simultaneously learns $T(z, t)$ and κ from the boundary and measurement data. Data constraints are applied at three locations: the surface ($z = 0$, RAD data), the mole ($z = 9.4$ cm, TEM-A data), and the deep boundary ($z = 40$ cm, fixed at $T = 225.6$ K). The PDE residual uses the rescaled form $(1/\kappa) \cdot \partial T / \partial t - \partial^2 T / \partial z^2 = 0$ to balance term magnitudes. Loss weights are $\lambda_{\text{BC}} = 100$, $\lambda_{\text{data}} = 50$.

Three initialisations were tested to probe the convergence basin: $\kappa_0 = 1 \times 10^{-9}$, 1×10^{-8} , and $3.93 \times 10^{-8} \text{ m}^2/\text{s}$. Each was run for 1.7–2.0 million iterations via sequential restarts with fresh Adam optimisers. Figure 5 shows the results.

All three trajectories are still drifting downward at the end. The drift decelerates over time, but the loss decreases monotonically across the observed range. After nearly 2 million iterations per trajectory, none has plateaued.

The root cause of the non-convergence can be diagnosed analytically, with no neural network involved. The 1D heat equation with periodic surface forcing has an exact solution at depth z :

$$T(z, t) = T_{\text{mean}} + A \exp(-z/\delta) \cos(\omega t - z/\delta), \quad \delta = \sqrt{\kappa P / \pi} \quad (6)$$

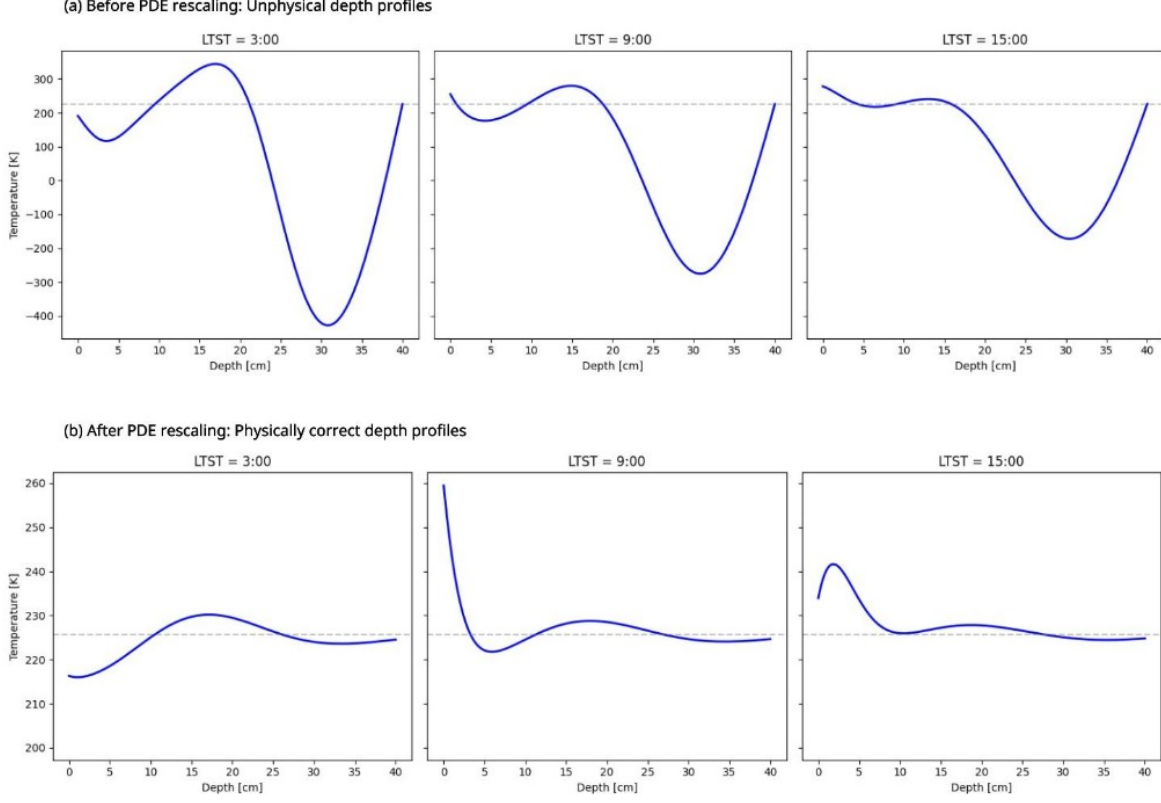


Figure 3: PINN-predicted temperature profiles $T(z)$ at three times of sol. (a) Before PDE rescaling the small effective diffusivity allows unphysical spatial oscillations reaching ± 400 K. (b) After dividing the PDE by κ_{eff} to balance both residual terms: profiles remain within the physically correct 200–260 K range and converge to $T_{\text{mean}} \approx 225.6$ K at depth. The dashed line marks T_{mean} .

This gives two independent observables at depth z : the amplitude ratio $\chi = \exp(-z/\delta)$ and the phase lag $\varphi = z/\delta$. Both are functions of a single unknown (δ , equivalently κ). For a true point sensor, they must yield the same δ . Using the Fourier-extracted observables from the TEM-A and RAD data ($\chi = 0.0595$, $\varphi = 1.2144$ rad):

Table 3: Amplitude- vs. phase-derived κ at the representative depth (9.4 cm).

Observable	δ (cm)	κ (m^2/s)	Ratio to Spohn
Amplitude $\chi = 0.0595$	3.33	3.93×10^{-8}	1.00×
Phase $\varphi = 1.214$ rad	7.74	2.12×10^{-7}	5.40×

The amplitude ratio and phase lag, interpreted as a point measurement at 9.4 cm, require κ values differing by 440%. Notably, the amplitude-derived κ matches Spohn et al.’s published value exactly. This is expected, since the representative depth of 9.4 cm was calibrated specifically to reproduce the first-harmonic amplitude ratio. The phase lag, however, is sensitive to the full waveform including higher harmonics, which the spatial averaging distorts differently for each frequency. This indicates that the mole signal is incompatible with a point-sensor measurement of undisturbed 1D diffusion. The inconsistency arises because the TEM-A physically averages the temperature over the full 35 cm mole length.

Neither making κ depth-dependent across three soil layers nor shifting the representative depth to the top of the mole (1.7 cm) resolved the inconsistency. Both modifications address the wrong problem.

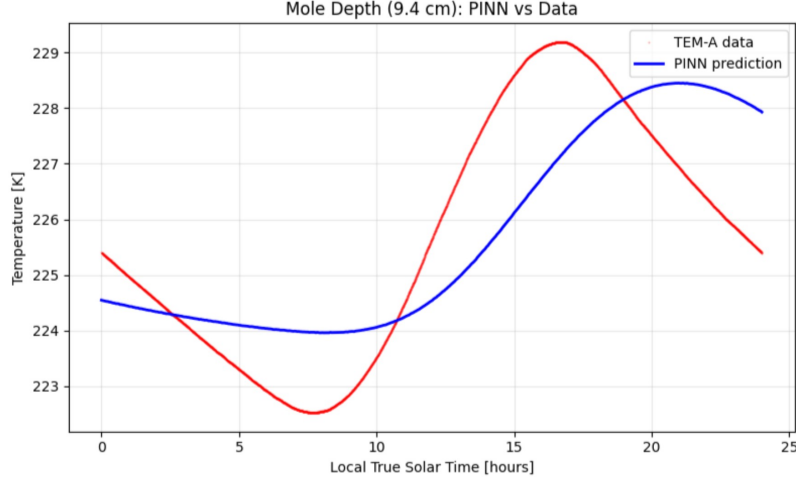


Figure 4: Forward PINN prediction at $z = 9.4$ cm compared with TEM-A data. The PINN reproduces amplitude mismatch and exhibits a systematic phase lag of approximately 2–3 hours.

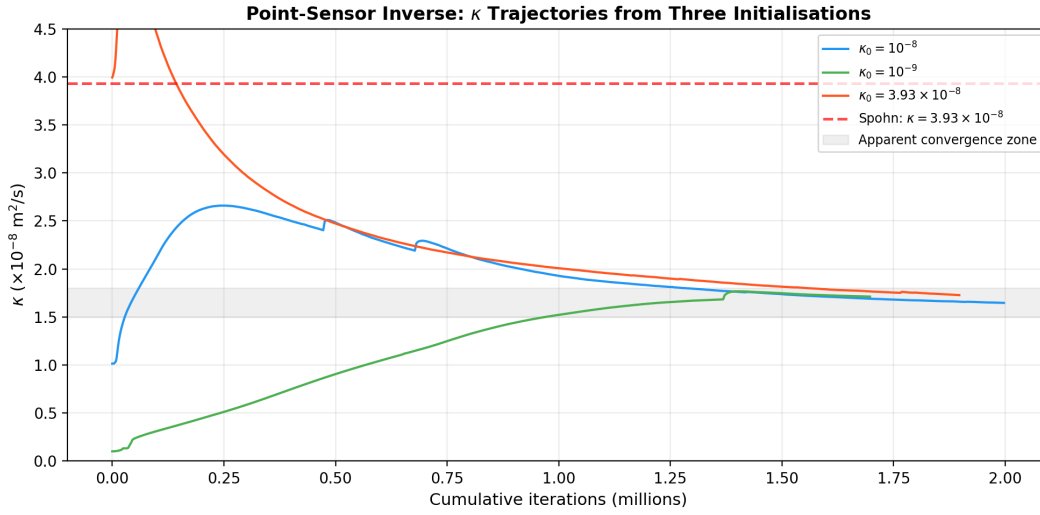


Figure 5: κ trajectories from three initialisations. All trajectories approach the $1.5\text{--}1.8 \times 10^{-8} \text{ m}^2/\text{s}$ region but remain in drift. The $\kappa_0 = 10^{-9}$ trajectory rises $17\times$ before turning over; the $\kappa_0 = 3.93 \times 10^{-8}$ trajectory falls monotonically.

The point-sensor model does not yield a well-defined inverse problem, as a single representative depth cannot capture the full waveform of the input signal; it is only valid for the first harmonic. The spatial averaging of the mole signal must be modelled explicitly.

4.3 Spatial Average inverse

As described in Section 2, the HP³ mole is a 40 cm aluminium cylinder, not a point sensor. Therefore the TEM-A sensor does not record the soil temperature at any single depth; it records the equilibrium temperature of the aluminium body. This equilibrium temperature is set by the condition that the net heat flux into the mole vanishes:

$$\int_{z_0}^{z_1} k_{\text{soil}} [T_{\text{soil}}(z, t) - T_{\text{mole}}(t)] dz = 0 \quad (7)$$

For approximately uniform k_{soil} , this gives:

$$T_{\text{mole}}(t) = \frac{1}{z_1 - z_0} \int_{z_0}^{z_1} T_{\text{soil}}(z, t) dz \quad (8)$$

Thus, the mole temperature is the spatial mean of the undisturbed soil over its depth extent.

Spohn et al. report a diurnal representative depth of 9.4 cm. This is the depth at which a hypothetical point sensor would reproduce the same first-harmonic amplitude ratio as the spatial average. Higher harmonics decay faster ($\delta_n = \delta_1/\sqrt{n}$), so no single depth captures the full waveform.

We implemented this problem by discretising the spatial average using $N = 100$ depth points uniformly spaced from z_0 to z_1 :

$$\hat{T}_{\text{mole}}(t_i) = \frac{1}{N} \sum_{j=1}^N T_{\text{net}}(z_j, t_i) \quad (9)$$

The data loss becomes:

$$\mathcal{L}_{\text{data}} = \frac{1}{M} \sum_{i=1}^M \left| \hat{T}_{\text{mole}}(t_i) - T_{\text{data}}(t_i) \right|^2 \quad (10)$$

where $M = 5,550$ is the number of TEM-A temperature measurements in the Sol 1202 record, retained after filtering for valid LTST and physical temperature range.

The boundary condition loss and physics loss remain unchanged. The only modification is replacing a single network evaluation per data point with an average over 100 evaluations.

The spatial average model achieved a mole signal RMSE of 0.39 K, with the persistent phase lag from the point-sensor model largely eliminated. The surface fit also improved (RMSE of 0.32 K), likely because the better-constrained subsurface temperature field feeds back into overall solution quality through the physics loss.

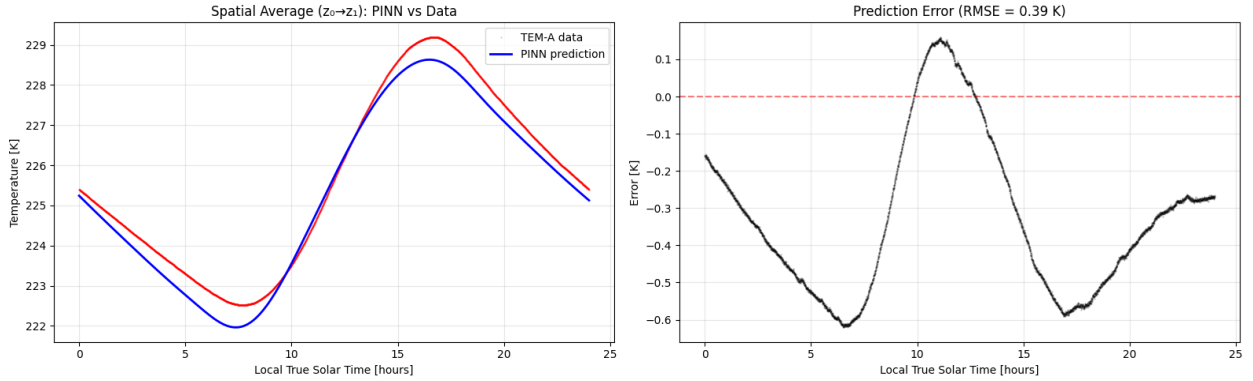


Figure 6: Spatial average forward PINN results. Top: predicted spatially averaged temperature over $[z_0, z_1]$ compared with TEM-A data, showing the elimination of the phase lag observed in the point-sensor model (cf. Figure 5). Bottom: prediction error over one sol, with RMSE = 0.39 K. The residual structure suggests a small remaining model–data mismatch rather than random noise.

While the fit quality improved substantially, the recovered κ diverged significantly from Spohn et al.’s published value. Three independent initialisations of the inverse PINN, spanning two orders of magnitude ($\kappa_0 \in \{1 \times 10^{-8}, 3.93 \times 10^{-8}, 1 \times 10^{-7}\} \text{ m}^2/\text{s}$), converged to a common value of $\kappa = 6.278 \pm 0.016 \times 10^{-8} \text{ m}^2/\text{s}$, with a cross-run spread of 0.49%. As shown in Figure 7, initialisations from below and above the final value bracket the solution from opposite directions, confirming that the recovered κ reflects a genuine basin of the loss landscape rather than an optimisation artifact. We note that κ stabilised while the physics residual continued to decline, indicating the network weights had not fully converged; however, the parameter estimate remained stable throughout this continued training.

The recovered $\kappa = 6.28 \times 10^{-8} \text{ m}^2/\text{s}$ is 59.7% above Spohn et al.’s published value of 3.93×10^{-8} . This discrepancy is not an optimisation failure; it is independently confirmed by the analytical solution of the spatial-average model, which yields $\kappa = 6.35 \times 10^{-8} \text{ m}^2/\text{s}$ from both amplitude-ratio and phase-lag methods (agreement within 1%).

The PINN is faithfully recovering the correct κ for the model it was given. The systematic inflation arises because the uniform spatial average ignores the mole’s role as a thermal fin, actively conducting heat from the warm shallow end to the cool deep end and boosting the observed signal amplitude beyond what passive averaging of the undisturbed temperature field would predict. Recovering Spohn’s value requires modelling this coupled mole–soil heat exchange, which is inherently a 2D axisymmetric problem.

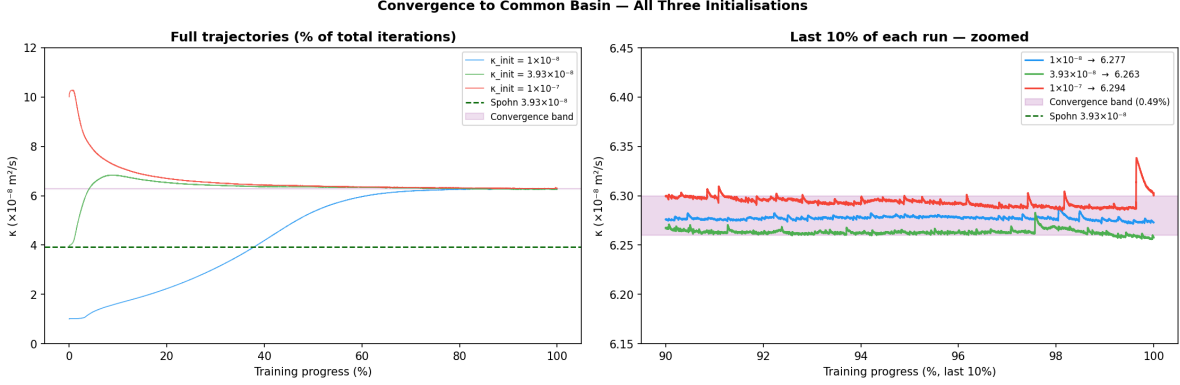


Figure 7: Left: all three trajectories normalised to training progress (%). Right: final 10% of each run zoomed to show convergence. Purple band marks the convergence region ($6.26\text{--}6.30 \times 10^{-8} \text{ m}^2/\text{s}$, width 0.49%).

Table 4: Multi-initialisation κ recovery — spatial average PINN.

κ_{init}	Total iters	Last-10% mean κ ($\times 10^{-8} \text{ m}^2/\text{s}$)	Std (% of mean)	Drift (% of mean)	vs. Grand mean
1×10^{-8}	350k	6.2772	0.0302%	−0.0488%	−0.01%
3.93×10^{-8}	900k	6.2631	0.0403%	−0.1488%	−0.24%
1×10^{-7}	800k	6.2941	0.0978%	+0.0130%	+0.25%
Grand mean	—	6.2781	—	—	—
Cross-run spread	—	0.0310 (0.49%)	—	—	—
Spohn et al.	—	3.9300	—	—	+37.4% (reference)

Table 5: Convergence comparison: spatial average (Model A) vs. point sensor (Model B).

Criterion	Spatial Average (Model A)	Point Sensor (Model B)
Multi-init spread	0.49%	20.8% (still narrowing)
κ plateaued?	Yes	No — still drifting
Analytical confirmation	PINN / Analytical = 0.99	No unique analytical target
χ - φ consistency	< 1%	440%
Loss minimum visible?	Yes (sharp)	No (monotonic decrease)
Physical interpretation	Valid model, quantifiable bias	Invalid forward model

4.4 2D Axisymmetric Model

The thermal fin effect identified in Section 4.3 requires modelling the radial heat exchange between the mole and surrounding soil. We extend the PINN to 2D axisymmetric coordinates.

We model the mole as a vertical cylinder (projecting the 30° tilt onto the vertical, following Spohn’s approach). Axisymmetry eliminates angular dependence. The PINN takes (r, z, t) as inputs and outputs $T(r, z, t)$, where r is the radial distance from the mole’s central axis.

The heat equation in cylindrical coordinates with axisymmetry:

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial z^2} \right) \quad (11)$$

The $\frac{1}{r} \cdot \frac{\partial T}{\partial r}$ term accounts for the unequal areas of cylindrical shell faces. No singularity issue arises because the domain starts at $r = R_{\text{mole}} \approx 1.35 \text{ cm}$ (mole radius), never at $r = 0$.

The domain extends radially from the mole surface ($R_{\text{mole}} \approx 1.35$ cm) to $r_{\text{max}} \approx 25$ cm (r_{max} is chosen at approximately 15–20 mole radii, where the thermal perturbation is negligible). The vertical and temporal domains are unchanged from the 1D setup ($z \in [0, 0.4]$ m, $t \in [0, 88775]$ s).

Four boundary conditions define the problem:

- **Surface** ($z = 0$ m): the RAD Sol 511 radiometer data, which provides the diurnal surface temperature forcing.
- **Deep bottom** ($z = 0.4$ m): fixed at $T = 225.6$ K, the mean temperature at depth where the diurnal signal has vanished.
- **Mole surface** ($r = R_{\text{mole}}$, $z \in [z_0, z_1]$): TEM-A Sol 1202 data applied as a Dirichlet BC; because $k_{\text{Al}} \gg k_{\text{soil}}$, the mole is isothermal at each instant, so every point along its length shares a single temperature $T_{\text{mole}}(t)$.
- **Far field** ($r = r_{\text{max}}$): a Neumann condition $\partial T / \partial r = 0$ enforces zero radial heat flux, encoding the assumption that the mole’s perturbation has fully decayed at this distance.

The forward PINN produced physically correct results: $T(r)$ decays monotonically from mole to far-field, heatmaps show diurnal penetration to ~ 0.05 m (consistent with skin depth), and $T(z)$ profiles separate near the surface and converge at depth. Total loss reached 3×10^{-5} . At the mole midpoint ($z = 0.19$ m), the 2D model reproduces the TEM-A signal with RMSE = 0.03 K (Figure 9), a 48 $\times$ improvement over the 1D point-sensor forward model (1.44 K) and 13 $\times$ over the spatial average (0.39 K). Because TEM-A is prescribed as a boundary condition, the low RMSE confirms that the PINN satisfies the imposed constraints rather than providing independent predictive validation. However, the physically plausible radial decay profile (Figure 8) indicates that the axisymmetric geometry captures the dominant physics of the mole–soil interaction.

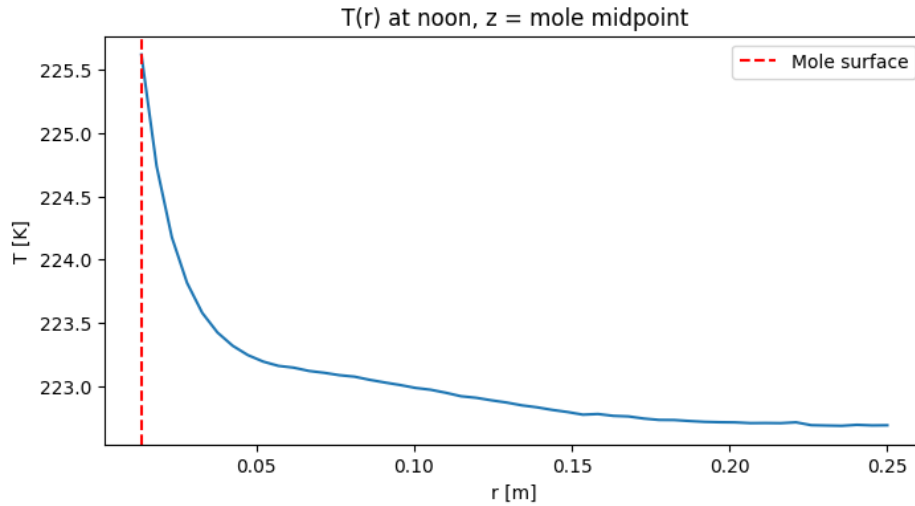


Figure 8: Radial temperature profile $T(r)$ at the mole midpoint depth at noon (LTST = 12:00), predicted by the 2D axisymmetric forward PINN. The dashed red line marks the mole surface ($R_{\text{mole}} = 1.35$ cm). Temperature decays sharply within the first ~ 5 cm from the hull and approaches the undisturbed far-field value by $r \approx 15$ cm, confirming that the mole’s thermal perturbation is localised.

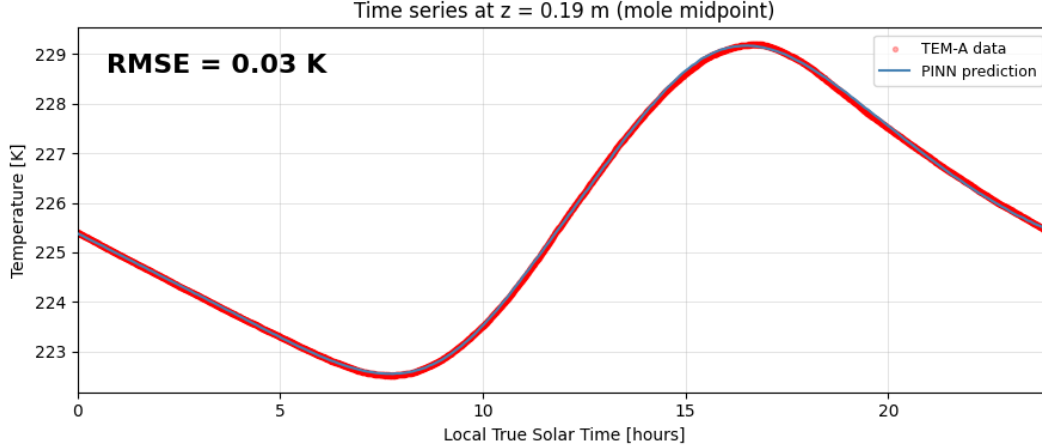


Figure 9: 2D axisymmetric forward PINN prediction at the mole midpoint ($z = 0.19$ m) compared with TEM-A data. RMSE = 0.03 K.

Several training strategies were attempted for the inverse problem: joint optimisation, warm-starting from the converged forward solution, and multi-phase training with frozen/unfrozen network weights. In all cases κ decayed to values below 1.7×10^{-9} , approximately $0.04 \times$ the true value. This reflects a structural identifiability failure: the network has sufficient freedom to reshape the interior temperature field to accommodate any κ , and the gradient-based optimiser exploits this by driving κ toward zero.

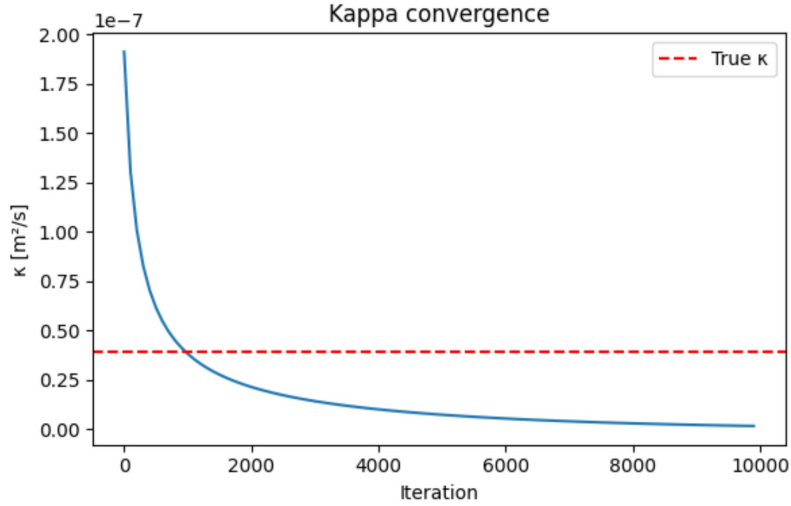


Figure 10: Representative κ trajectory for the 2D axisymmetric inverse PINN. The recovered diffusivity decays monotonically past the Spohn et al. reference (dashed line) toward zero, indicating failure to converge. The dashed red line marks Spohn's $\kappa = 3.93 \times 10^{-8}$.

A perturbation decomposition approach was attempted. It consisted of splitting the resulting temperature field into two components:

$$T(r, z, t) = T_{1D}(z, t; \kappa) + T_{\text{pert}}(r, z, t) \quad (12)$$

The decomposition was designed to constrain κ by forcing it to appear in both the analytical background solution and the perturbation PDE, creating tension that should pin its value. However, the results did not change: κ drifted downwards in every configuration tried. The same co-adaptation failure observed in the direct approach persisted: the network reshaped the perturbation field to accommodate decreasing κ . The optimiser exploits this by pushing $\kappa \rightarrow 0$ to trivially satisfy the PDE.

For this class of thermal inverse problem, interior observation data appears necessary to break the co-adaptation between network weights and the learnable parameter. The 2D setup only had data on boundaries. There are no measured temperatures inside the soil volume between the mole and the far field. Without interior data to pin the temperature field, the PINN adjusts its spatial structure to accommodate whatever κ the optimiser tries, and the loss decreases regardless of whether κ is physically correct.

Table 6: Summary of all model variants, recovered κ values, and fit quality. Converged models show cross-run spread; unconverged models are marked with the observed behaviour.

Model	Observation operator	κ recovered ($\times 10^{-8}$ m ² /s)	Converged?	RMSE _{mole} (K)	Key finding
Forward (κ fixed)	Point at $z = 9.4$ cm	3.93 (fixed)	—	1.44	Phase lag + amplitude mismatch; rescaling improvement
Point-sensor inverse	Point at $z = 9.4$ cm	N/A (drifting)	No	—	440% χ - φ inconsistency; ill-posed
Spatial avg inverse	$\int T dz$ over $[z_0, z_1]$	6.28 ± 0.016	Yes (0.49%)	0.39	60% above Spohn; fin effect
2D axisymmetric forward	$T_{\text{mole}}(t)$ at $r = R_{\text{mole}}$	3.93 (fixed)	—	0.03	Physically correct $T(r)$ decay
2D axisymmetric inverse	$T_{\text{mole}}(t)$ at $r = R_{\text{mole}}$	$\rightarrow 0$	No	—	Structural identifiability failure

5 Discussion

5.1 The Observation Operator

The central finding of this work is that inverse PINNs can converge reliably to a physically consistent but incorrect parameter value when the forward model does not match the true measurement process. When the model is more severely misspecified, the recovered parameter may lose physical meaning entirely.

The spatial average model (Section 4.3) and point-sensor model (Section 4.2) demonstrate two forms of this failure at different levels of severity. In the point-sensor formulation, the forward model is internally inconsistent: the amplitude ratio and phase lag demand κ values differing by 440%, and the PINN drifts toward a κ that has no physical meaning. In the spatial average formulation, the forward model is self-consistent, as the PINN and analytical solution agree within 1%. However, the recovered $\kappa = 6.28 \times 10^{-8}$ m²/s is 60% above Spohn et al.’s value because the uniform average omits the mole’s thermal fin effect. The PINN is faithfully solving the problem it was given; the discrepancy lies in the problem definition, not the solver.

In the inverse problems literature, the mapping from the physical field to the sensor reading is known as the observation operator (Aster et al., 2018). Recent work has shown that the choice of observation operator can significantly affect posterior estimates even in well-posed inverse problems (Cvetković et al., 2023). Our results suggest that for real-instrument inverse problems, the fidelity of this operator matters more than network architecture, loss weighting, learning rate schedules, or any other machine learning consideration. All of the standard PINN improvements we attempted (Fourier features, residual rescaling, weight rebalancing, increased network capacity, and learning rate scheduling) were unable to compensate for a misspecified observation operator.

5.2 Structural identifiability

The 2D axisymmetric experiments revealed a second limitation: even when the measurement model is correct in principle, boundary-only data may be insufficient to uniquely determine the learnable parameter. In the 2D formulation, the network has enough representational freedom to reshape the interior temperature field to accommodate any κ , and the gradient-based optimiser exploits this by driving κ toward zero. A perturbation decomposition ($T = T_{\text{ID}} + T_{\text{pert}}$) was designed to constrain κ by forcing it to appear in both the analytical background and the perturbation PDE, but in practice the network absorbed the inconsistency. This co-adaptation between network parameters and the learnable physical constant echoes the broader finding that PINN training failures typically stem from optimisation pathologies rather than insufficient representational capacity (Krishnapriyan et al., 2021; Wang et al., 2022).

The common thread across all inverse configurations is the absence of interior observation data. Without measured temperatures inside the domain, the network’s spatial structure is underdetermined, and a co-adaptation between

network weights and κ becomes the path of least resistance for the optimiser. This precisely violates the uniqueness criterion of a well-posed problem as stated by Hadamard (1923). Woodbury et al. (2023) identify this incomplete spatial information as a structural feature of sparse-sensor IHCPs rather than an artifact specific to neural network optimisation. Classical methods do not eliminate this ambiguity either; they instead rely on additional sensor placement or explicit regularisation to select a single, stable estimate from an otherwise non-unique solution space.

5.3 Practical Lessons

For practitioners applying inverse PINNs to real instrument data, our experience suggests five priorities:

First, check whether the observation operator is analytically consistent with the measurement data. If the sensor signal can be decomposed into independent observables (e.g. amplitude and phase), each should yield the same parameter value. In our case, we performed this analytical consistency check after training the inverse point-sensor model. Had we performed it before training, the 440% inconsistency between amplitude-derived and phase-derived κ would have been apparent without a single iteration of training.

Second, validate the forward problem before attempting parameter recovery. The forward problem should reproduce the sensor signal with physically correct depth profiles, not merely low RMSE at measurement locations. Our 4×128 network achieved RMSE of 0.006 K while producing temperature swings of ± 400 K at depth. This is a cautionary example that low data-space error does not guarantee physical correctness.

Third, verify PINN-recovered parameters against analytical solutions where possible. The 1% agreement between our PINN and analytical κ values confirmed that the PINN was functioning correctly and isolated the discrepancy as a modelling issue rather than an optimisation failure.

Fourth, test convergence from multiple initialisations. A single run that appears to converge may simply be a snapshot of a drifting trajectory, as we observed with the point-sensor inverse problem.

Fifth, when convergence fails despite a correctly specified observation operator, the fix is unlikely to lie in network architecture, training strategy, or loss weighting. Classical methods such as Tikhonov regularisation provide an explicit, tunable parameter to trade solution stability against fit; the physics residual offers no equivalent dial. In our 2D axisymmetric experiments, no combination of training strategies recovered a valid κ , because the limitation was insufficient measurement information rather than an optimisation deficiency. As in classical IHCP, the correct response is to add measurement information (e.g. interior sensors) or impose an explicit regularisation constraint, rather than attempting to compensate for missing physical information through architectural changes.

5.4 Limitations

Several limitations of this work should be noted. The physics residual continued declining in all spatial-average runs after κ had stabilised, indicating that the network weights had not fully converged even though the parameter estimate had. The 1×10^{-7} initialisation showed the highest variability among the three runs and should be interpreted as a bracketing constraint rather than independent confirmation of convergence. Additionally, the RAD surface temperature data (Sol 511) and TEM-A mole data (Sol 1202) are from different Mars years; while selected to match seasonal conditions, small differences in atmospheric opacity or surface albedo may introduce systematic error.

Furthermore, this work assumes spatially uniform thermal diffusivity. Soil layering or variations in moisture content could in principle contribute to the discrepancy with Spohn et al.’s value, though Spohn et al. (2024) make the same homogeneity assumption in their Fourier analysis.

To assess sensitivity to measurement noise, three additional inverse runs were performed with ± 0.5 K Gaussian noise added to the TEM-A data (approximately 7% of the diurnal signal amplitude). The recovered κ shifted by at most 0.85% from the clean-data result, comparable to the cross-initialisation spread of 0.49%. This confirms that the recovered value is insensitive to realistic sensor noise levels.

6 Conclusion

Throughout this investigation we tested the ability of PINNs to retrieve valid physical results from real world noisy data. A progression of models was developed to recover the thermal diffusivity of Martian subsurface soil from InSight HP³ data, each addressing a limitation of its predecessor.

The spatial average model recovered $\kappa = 6.28 \times 10^{-8} \text{ m}^2/\text{s}$ with robust convergence across three initialisations (0.49% spread), independently confirmed by the analytical solution (6.35×10^{-8} , agreement within 1%). The 60% discrepancy

with Spohn is consistent with the mole’s thermal fin effect, which the uniform spatial average does not capture. The point-sensor model revealed a more fundamental failure: the amplitude ratio and phase lag of the mole signal demand κ values differing by 440% when interpreted at a single depth, rendering the inverse problem ill-posed.

This shows that when using PINNs with real sensor data, having a correctly defined observation operator is more important than using the appropriate ML tricks. If the wrong observation operator is used, the PINN can converge, but to a potentially wrong value.

In order to improve our results in the future, two directions may resolve the remaining discrepancy. First, adding interior sensors in future mission design would provide the internal constraints needed to break the network-parameter co-adaptation observed in the 2D models. The second potential lead is to complete a full numerical simulation of the coupled mole system. This will quantify whether the fin effect fully explains the 60% gap produced between Spohn’s value and the spatial average result.

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